

Unit 1

Theory of Probability

Introduction

- Probability means possibility.
- It is a branch of mathematics that deals with the occurrence of a random event.
- The value is expressed from zero to one.
- Probability has been introduced in Mathematics to predict how likely events are to happen.
- The meaning of probability is basically the extent to which something is likely to happen. This is the basic probability theory, which is also used in the probability distribution, where you will learn the possibility of outcomes for a random experiment.
- To find the probability of a single event to occur, first, we should know the total number of possible outcomes.

Probability Terms , Definitions and Example

Term	Definition	Example
Experiment or Trial	A series of actions where the outcomes are always uncertain	1. The tossing of a coin. 2. Selecting a card from a deck of cards. 3. Throwing a dice.
Event	It is a single outcome of an experiment	Getting a Heads while tossing a coin is an event
Mutually Exclusive Event	Two Events are said to be mutually exclusive events if both cannot happen simultaneously	1. When a coin is tossed we cannot get Heads(H) and Tails(T) both simultaneously. 2. If we draw a card from pack of, let us define getting an heart as event A and getting a king card as event B, then A and B are NOT mutually exclusive events, as we may get a single card as "heart" and "king" i.e "king of hearts"

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Term	Definition	Example
Equally likely events	Events are said to be equally likely if we have no reason to expect one particular event to occur in preference to any other.	In a toss of a coin getting a “head” and “tail” are equally likely events.
Exhaustive Events	A group of events is said to be exhaustive if it includes all possible events.	1. In a toss of coin there are only TWO events H and T. Hence group of {H,T} is exhaustive.
Sample Space	The set of all the possible outcomes to occur in any trial	1. Tossing a coin, Sample Space $(S) = \{H,T\}$. 2. Rolling a die, Sample Space $(S) = \{1,2,3,4,5,6\}$

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Term	Definition	Example
Sample Point	It is one of the possible results	In a deck of Cards: • 4 of hearts is a sample point. • The queen of clubs is a sample point
Outcome	Possible result of a trial/experiment	Standard 52-card deck, A = Draw a heart, then A' = Don't draw a heart
Complimentary event	The non-happening events. The complement of an event A is the event, not A (or A')	1. Tossing a coin, Sample Space (S) = {H,T}. 2. Rolling a die, Sample Space (S) = {1,2,3,4,5,6}
Impossible Event	The event cannot happen	In tossing a coin, impossible to get both head and tail at the same time

- **Depending on the number of outcomes are classified as :**

- a. Deterministic Experiment
- b. Random or Non-Deterministic Experiment

- **Deterministic Experiment:**

- The experiment in which outcome is always unique and Hence certain. Thus the outcome can be predicted before the trial of an experiment

- E.g.

- Turning on light switch
- Throwing a ball in the sky
- $V=d/t$

- **Non-Deterministic Experiment:**

- A non-deterministic experiment or a random experiment is the experiment in which there are more than one possible outcomes. Due to uncertainty outcomes of this experiment cannot be predicted before performing the experiment

- E.g.

- Rolling a die
- Student appearing for an exam
- Blood Group in a bank

A box contains 20 tickets numbered 1 to 20. One ticket is drawn at random from the box. Find the probability that the number on the ticket drawn is

- i. Greater than 12
- ii. Even number
- iii. Multiple of 3
- iv. Divisible by 3 or 4
- v. Divisible by 3 and 4

Two fair dice are rolled Find the probability that the Sum of number on uppermost faces is

- i. 8
- ii. Less than 5
- iii. A perfect square
- iv. Multiple of 4

Q.1 A box contains 7 white 5 black and 4 red balls. 2 balls are drawn at random. Find the probability that

- i. Both balls are white
- ii. One is black and other is red

Q.2 A committee of 4 is to be formed from 3 engineers, 3 economists, 2 statisticians and 1 C.A. What is the probability that

- a. Each of the 4 categories of 4 profession is included in the committee
- b. Committee consists of C.A and atleast one engineer

Q.3 A committee of 4 persons is to be formed from among 5 men and 5 women. Find the prob that committee will consist atleast two women

Types of Probability:

- Theoretical / Classical Probability
- Empirical / Experimental Probability
- Subjective Probability

Theoretical Probability / Classical Probability/Apriori Probability

If an experiment results in n mutually exclusive exhaustive and equally likely outcomes out of which m are favouring to event A then probability of occurrence of event is given by

$$P(\text{Event } A) = \frac{\text{No of favourable outcomes}}{\text{Total number of possible outcomes}}$$

$$P(A) = \frac{m}{n}$$

Empirical / Experimental Probability:

If the outcomes of experiment are not equally likely then it is possible to conduct experiment a large number of times to find estimate of probability and then the estimated prob from the results is referred to as Empirical probability

$$\text{Experimental Probability} = \frac{\text{Number of times an event occurs}}{\text{Total number of trials}}$$

Subjective Probability

Subjective probability is the opinion or probable outcome based on an individual's perspective, personal experience, and sense of judgment. It is entirely based on an individual's belief system so it differs from person to person, and there is no set pattern to derive it.

3. Axiomatic Probability:

In axiomatic probability, a set of rules or axioms are set which applies to all types. These axioms are set by Kolmogorov and are known as Kolmogorov's three axioms. With the axiomatic approach to probability, the chances of occurrence or non-occurrence of the events can be quantified. The axiomatic probability lesson covers this concept in detail with Kolmogorov's three rules (axioms).

Axioms:

1. $P(A) \geq 0$:

This axioms states that the probability of any event is greater than zero, which means the **probability of any event cannot be negative.**

2. $P(S) = 1$:

This axioms states that the **sum of all probabilities is equal to 1.**

3. $P(A \cup B) = P(A) + P(B)$:

This axioms states that if **two events are mutually exclusive** than the **probability that either of them will occur is equal to the sum of their probabilities.**

Laws and Theorems of Probability

1. $P(S) = 1$ -----Sum of all probabilities

2. $P(\Phi) = 0$ ----- Φ denotes impossible event

3. $P(\bar{A}) = 1 - P(A)$ -----Complementary event

4. Addition theorem for Two events is given as;

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Addition theorem for Three events(A ,B and C) is given as;

$$P(A \cup B \cup C)$$

$$= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

5. $P(A \cap B)$ ----Multiplicative Theorem indicates both events A and B together.

What is the chance that a leap year selected at random will contain 53 Sundays.

In the play of two fair dice a person loses if the sum obtained is 2 or 4 or 12. He wins if the sum is 5 or 11. Find the ratio of probability of losing to winning in the first throw.

In a town of 10000 families it was found that 40% of families buy newspaper A, 20% family buy newspaper B, 10% family buy newspaper C, 5% family buy newspaper A and B, 3% family buy newspaper B and C and 4% family buy newspaper A and C. If 2% family buy all the newspaper. Find the number of families which buy

1. Number of families which buy all three newspapers.
2. Number of families which buy newspaper A only, B only and C only
3. Number of families which buy None of A, B, C
4. Number of families which buy exactly only one newspaper
5. Number of families which buy newspaper A and B only
6. Number of families which buy newspaper B and C only
7. Number of families which buy newspaper C and A only
8. Number of families which buy at least two newspapers
9. Number of families which buy at most two newspapers
10. Number of families which buy exactly two newspapers

Conditional Probability

The probability of an event occurring given that another event has already occurred is called conditional probability

E.g. :

- a. Weather Forecast
- b. Shopping
- c. Smart Phone Usage

Conditional Probability Formula

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

Probability of A given B

Probability of A and B

Probability of B

Weather Forecast

Event A : Its raining

Event B: Sky is cloudy

Shopping:

Event A: Buying a product online.

Event B: Adding the product to the shopping cart.

Smart Phone Usage

Event A: Battery running out.

Event B: Using the phone heavily for gaming.

Example

1. Two fair dice is rolled If the sum 10 has appeared Find the Probability that one of the dice show
 - a. Number 4
 - b. Number 3

2. A family has two children. Find the probability that both children are girls if it is known that
 - a. One of the child is girl
 - b. The older child is a girl

3. Two dice are thrown simultaneously, and the sum of the numbers obtained is found to be 7. What is the probability that the number 3 has appeared at least once?

4. Sapan took part in two games. The probability of her passing both games is 0.4. The probability of her passing the first game is 0.6. What is the probability of her passing the second game given that she has passed the first game?

5. A group of 100 computer buyers, 40 bought CPU, 30 purchased monitor, and 20 purchased CPU and monitors. If a computer buyer chose at random and bought a CPU, what is the probability they also bought a Monitor?

Multiplication Law of Probability

- The multiplication rule of probability explains the condition between two events.
- For two events A and B associated with a sample space S set $(A \cap B)$ denotes the events in which both events A and event B have occurred.
- Hence, $(A \cap B)$ denotes the simultaneous occurrence of events A and B.
- Event $A \cap B$ can be written as AB.
- The probability of event AB is obtained by using the properties of conditional probability.
- According to the multiplication rule of probability, the probability of occurrence of both the events A and B is equal to the product of the probability of B occurring and the conditional probability that event A occurring given that event B occurs.

- If A and B are dependent events, then the probability of both events occurring simultaneously is given by:

$$P(A \cap B) = P(B) \cdot P(A|B) \text{ if } P(B) \neq 0$$

$$P(A \cap B) = P(A) \cdot P(B|A) \text{ if } P(A) \neq 0$$

- If A and B are two independent events in an experiment, then the probability of both events occurring simultaneously is given by:

$$P(A \cap B) = P(A) \cdot P(B)$$

Example

An urn contains 20 red and 10 blue balls. Two balls are drawn from a bag one after the other without replacement. What is the probability that both the balls are drawn are red?

Solution:

Let A and B denote the events that the first and the second balls are drawn are red balls. We have to find $P(A \cap B)$ or $P(AB)$.

$$P(A) = P(\text{red balls in first draw}) = 20/30$$

Now, only 19 red balls and 10 blue balls are left in the bag. The probability of drawing a red ball in the second draw too is an example of conditional probability where the drawing of the second ball depends on the drawing of the first ball.

Hence Conditional probability of B on A will be,

$$P(B|A) = 19/29$$

By multiplication rule of probability,

$$P(A \cap B) = P(A) \times P(B|A)$$

$$P(A \cap B) = \frac{20}{30} \times \frac{19}{29} = \frac{38}{87}$$

Take Home Task

1. You have a cowboy hat, a top hat, and an Indonesian hat called a songkok. You also have four shirts: white, black, green, and pink. If you choose one hat and one shirt at random, what is the probability that you choose the songkok and the black shirt?
2. Suppose you take out two cards from a standard pack of cards one after another, without replacing the first card. What is probability that the first card is the ace of spades, and the second card is a heart?

3.2 persons A and B appear for two vacancies in the same post. The probability of A's selection is $\frac{1}{6}$ and that of B's selection is $\frac{1}{5}$. What is the prob that

- a. Both will be selected
- b. Only one of them will be selected
- c. None of them will be selected

4. A lot contains 15 items out of which 3 are defective. Two items are drawn at random one after the other without replacement. Find the probability that

- a. All the items are defective
- b. First item is defective
- c. Only one item is defective

Independent events :

- Those events whose occurrence is not dependent on any other event.
- For example, if we flip a coin in the air and get the outcome as Head, then again if we flip the coin but this time we get the outcome as Tail.
- In both cases, the occurrence of both events is independent of each other.

Question: Let X and Y are two independent events such that $P(X) = 0.3$ and $P(Y) = 0.7$. Find $P(X \text{ and } Y)$, $P(X \text{ or } Y)$, $P(Y \text{ not } X)$, and $P(\text{neither } X \text{ nor } Y)$.

Solution:

Given $P(X) = 0.3$ and $P(Y) = 0.7$ and events X and Y are independent of each other.

$$P(X \text{ and } Y) = P(X \cap Y) = P(X) P(Y) = 0.3 \times 0.7 = 0.21$$

$$P(X \text{ or } Y) = P(X \cup Y) = P(X) + P(Y) - P(X \cap Y) = 0.3 + 0.7 - 0.21 = 0.79$$

$$P(Y \text{ not } X) = P(Y \cap X') = P(Y) - P(X \cap Y) = 0.7 - 0.21 = 0.49$$

$$\text{And } P(\text{neither } X \text{ nor } Y) = P(X' \cap Y') = 1 - P(X \cup Y) = 1 - 0.79 = 0.21$$

Mutually Exclusive Events :

Two events A and B are said to be mutually exclusive events if they cannot occur at the same time. Mutually exclusive events never have an outcome in common.

Bayes' Theorem

- It describes the probability of occurrence of an event related to any condition. It is also considered for the case of conditional probability.
- Bayes theorem is also known as the formula for the probability of “causes”.
- For example: if we have to calculate the probability of taking a blue ball from the second bag out of three different bags of balls, where each bag contains three different colour balls viz. red, blue, black.
- In this case, the probability of occurrence of an event is calculated depending on other conditions is known as conditional probability.

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Bayes theorem Applications

- One of the many applications of Bayes' theorem is Bayesian inference, a particular approach to statistical inference.
- Bayesian inference has found application in various activities, including medicine, science, philosophy, engineering, sports, law, etc.
- For example, we can use Bayes' theorem to define the accuracy of medical test results by considering how likely any given person is to have a disease and the test's overall accuracy.
- Bayes' theorem relies on consolidating prior probability distributions to generate posterior probabilities. In Bayesian statistical inference, prior probability is the probability of an event before new data is collected.

probability a hypothesis is true
given the evidence

probability a hypothesis is true
(before any evidence is present)

probability of seeing the evidence
if the hypothesis is true

$$P(\textcolor{red}{H}/\textcolor{green}{E}) = \frac{P(\textcolor{red}{H}) P(\textcolor{green}{E}/\textcolor{red}{H})}{P(\textcolor{green}{E})}$$

probability of observing the evidence

$P(H)$ = how likely A happens(Prior knowledge)- The probability of a hypothesis is true before any evidence is present.

$P(E)$ = how likely B happens(Marginalization)- The probability of observing the evidence.

$P(H|E)$ = how likely A happens given that B has happened(Posterior)-The probability of a hypothesis is true given the evidence.

$P(E|H)$ = how likely B happens given that A has happened(Likelihood)- The probability of seeing the evidence if the hypothesis is true.

A bag I contains 4 white and 6 black balls while another Bag II contains 4 white and 3 black balls. One ball is drawn at random from one of the bags, and it is found to be black. Find the probability that it was drawn from Bag I.

Let E_1 be the event of choosing bag I, E_2 the event of choosing bag II, and A be the event of drawing a black ball.

$$\text{Then, } P(E_1) = P(E_2) = \frac{1}{2}$$

$$\text{Also, } P(A|E_1) = P(\text{drawing a black ball from Bag I}) = \frac{6}{10} = \frac{3}{5}$$

$$P(A|E_2) = P(\text{drawing a black ball from Bag II}) = \frac{3}{7}$$

By using Bayes' theorem, the probability of drawing a black ball from bag I out of two bags,

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$$

$$= \frac{\frac{1}{2} \times \frac{3}{5}}{\frac{1}{2} \times \frac{3}{5} + \frac{1}{2} \times \frac{3}{7}} = \frac{7}{12}$$

A man is known to speak the truth 2 out of 3 times. He throws a die and reports that the number obtained is a four. Find the probability that the number obtained is actually a four.

Let A be the event that the man reports that number four is obtained.

Let E_1 be the event that four is obtained and E_2 be its complementary event.

Then, $P(E_1)$ = Probability that four occurs = $\frac{1}{6}$

$P(E_2)$ = Probability that four does not occur = $1 - P(E_1) = 1 - \frac{1}{6} = \frac{5}{6}$

Also, $P(A|E_1)$ = Probability that man reports four and it is actually a four = $\frac{2}{3}$

$P(A|E_2)$ = Probability that man reports four and it is not a four = $\frac{1}{3}$

By using Bayes' theorem, probability that number obtained is actually a four,

$$P(E_1|A) = \frac{P(E_1)P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)} = \frac{\frac{1}{6} \times \frac{2}{3}}{\frac{1}{6} \times \frac{2}{3} + \frac{5}{6} \times \frac{1}{3}} = \frac{2}{7}$$

- i. For HSC exam it was considered that probability of students scoring 650 and more marks as 0.25. A survey was conducted after the exam where students were asked question that whether they had taken any preparation class before the exam, the result of the survey was that probability of student taking preparation class scoring 650 or more marks was 0.6 and probability of students who did take preparation class yet scored less than 650 was 0.31. Calculate posterior probabilities.

Event A: Students scoring 650 and more

Event B: Students scoring less than 650

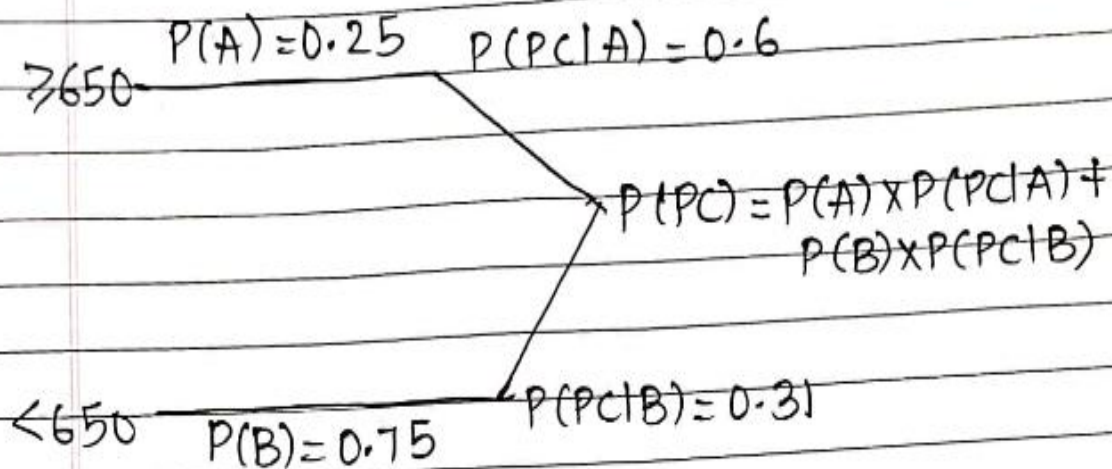
Event PC: Students taken any preparation class

Event ^{NPC} D: Students did not take any preparation class

$$P(A) = 0.25 \quad P(B) = 1 - 0.25 = 0.75$$

$$P(PC|A) = 0.6 \quad P(PC|B) = 0.31$$

a) $P(\geq 650 | PC)$



$$P(\geq 650 | PC) = \frac{P(PC | \geq 650) \times P(\geq 650)}{P(PC)}$$

$$= \frac{P(PC|A) \times P(A)}{P(PC)}$$

$$= \frac{0.6 \times 0.25}{0.3825}$$

$$= 0.3921568$$

b) $P(< 650 | PC)$

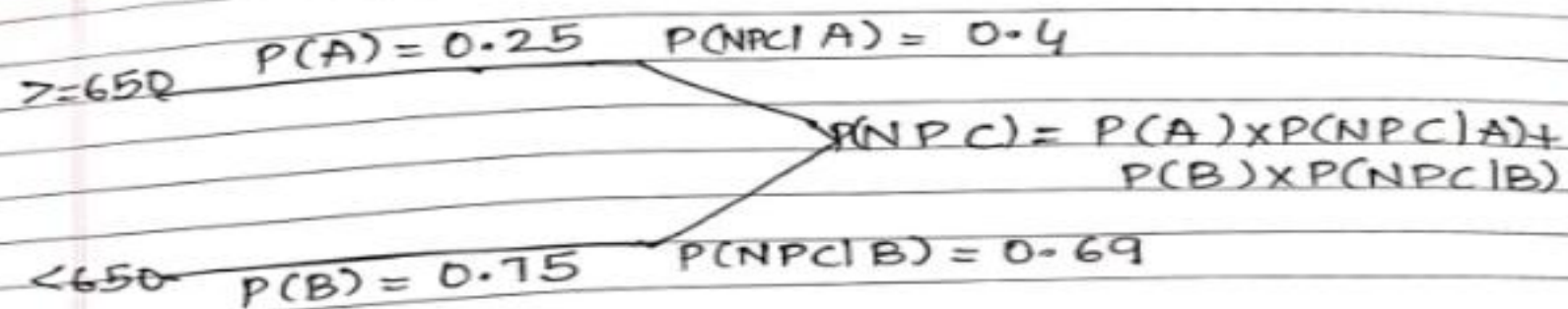
$$= \frac{P(PC | < 650) \times P(< 650)}{P(PC)}$$

$$= \frac{P(PC|B) \times P(B)}{P(PC)}$$

$$= \frac{0.31 \times 0.75}{0.3825}$$

$$= 0.3825$$

$$c) P(>=650|NPC) =$$



$$P(>=650|NPC) = \frac{P(NPC|\geq 650) \times P(\geq 650)}{P(NPC)}$$

$$= \frac{P(NPC|A) \times P(A)}{P(NPC)}$$

$$= \frac{0.4 \times 0.25}{0.6175}$$

$$= 0.1619$$

$$d) P(<650|NPC) = \frac{P(NPC|<650) \times P(<650)}{P(NPC)}$$

$$= \frac{0.69 \times 0.75}{0.6175}$$

Mathematical Tools for Probability

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graph TD; A[Mathematical Tools] --> B[Venn diagrams]; A --> C[Tree diagrams]; A --> D[Two-way contingency tables];
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Mathematical
Tools

Venn
diagrams

Tree diagrams

Two-way
contingency
tables

Venn diagrams

- It is used to show how events are related to one another.
- A Venn diagram can be very helpful when doing calculations with probabilities.
- In a Venn diagram each event is represented by a shape, often a circle or a rectangle.



- The region inside the shape represents the outcomes included in the event and the region outside the shape represents the outcomes that are not in the event.

Make a Venn Diagram for the following situation:

30 students were asked which sports they play.

- 20 play basketball in total
- 16 play football in total
- 15 play tennis in total
- 10 play basketball and tennis
- 11 play basketball and football
- 9 play football and tennis
- 7 play all three

Example (Solve using Venn diagram)

Q.1 There are 200 boys in Grade 12 at Marist Brothers High School. Their participation in sport can be broken down as follows:

107 play rugby

90 play soccer

63 play cricket

35 play rugby and soccer

23 play rugby and cricket

15 play rugby, soccer and cricket

190 boys play rugby, soccer or cricket

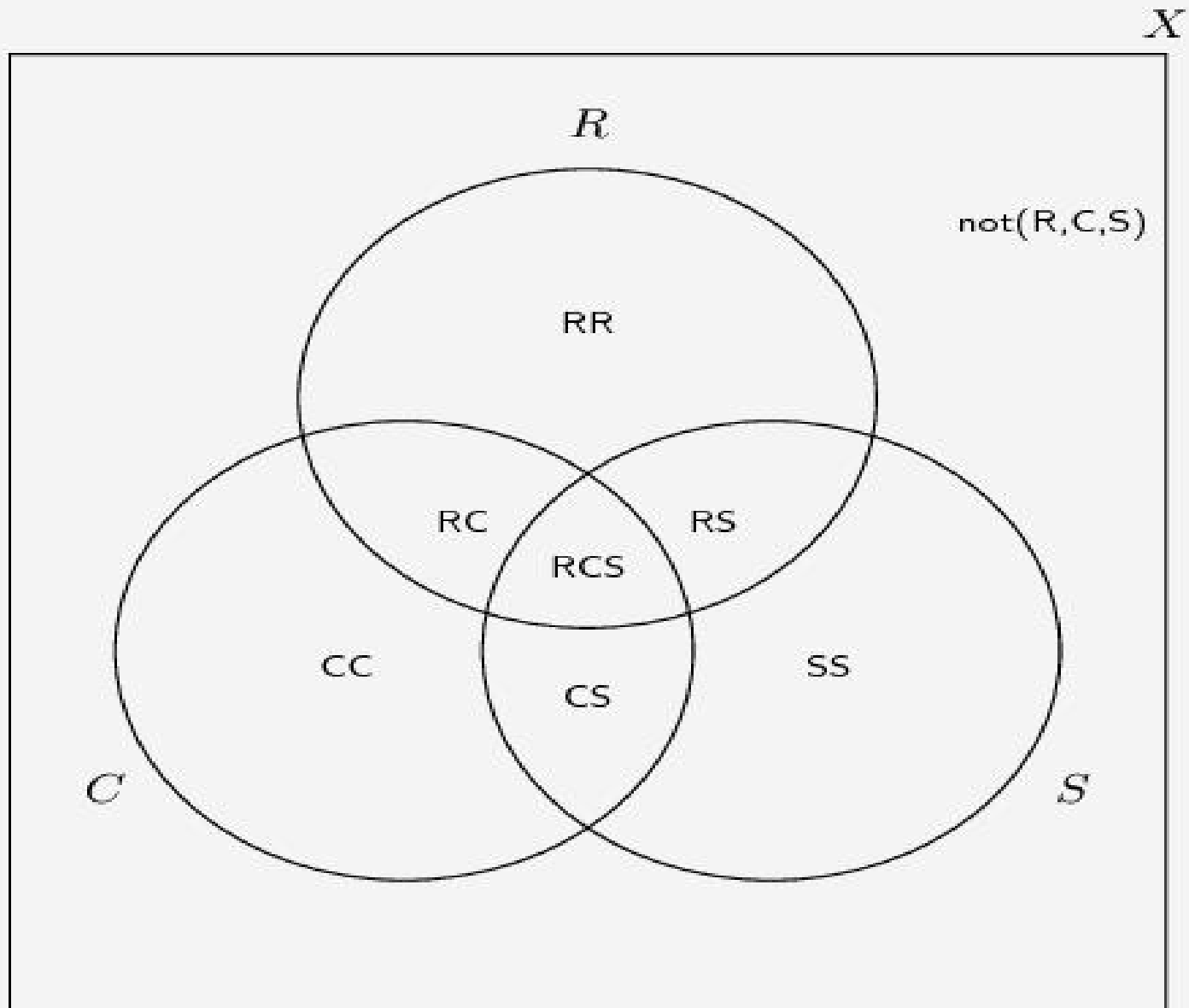
1. How many boys do not play any of these sports?

2. Draw a Venn diagram to illustrate the given information and use it to answer the following questions:

a. How many boys play soccer and cricket, but not rugby?

b. What is the probability that a randomly chosen Grade 12 boy at Marist Brothers High School will take part in at least two of the sports: rugby, soccer or cricket? Give your answer correct to 3 decimal places.

Calculate the counts for the different groupings



Solution

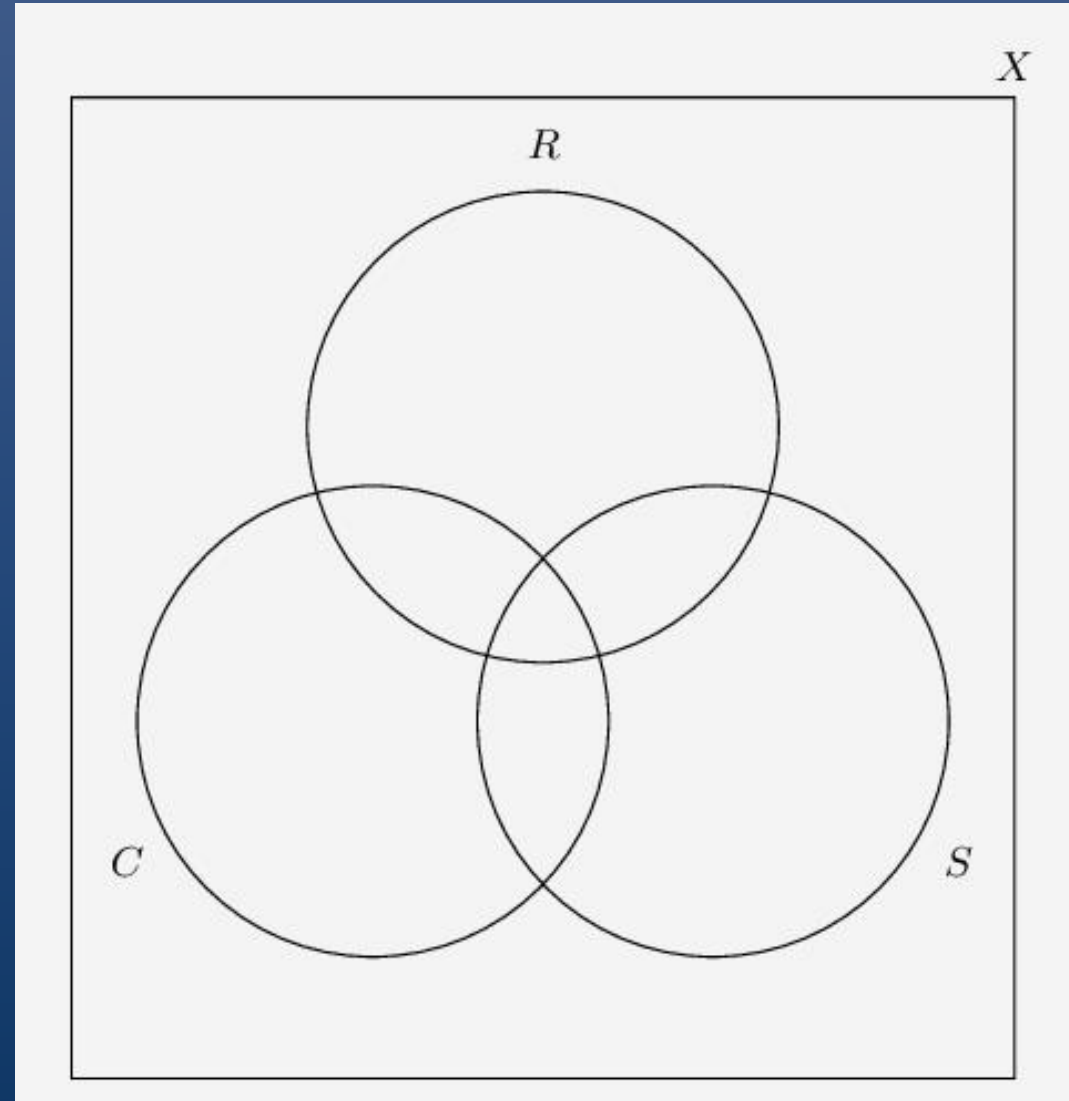
1. In order to calculate the number of boys playing none of the sports, we subtract the number of boys playing any of the three sports from the total number of boys in the sample space.

Not rugby, cricket or soccer
 $= 200 - 190 = 10$

Therefore 10 boys do not play rugby, cricket or soccer.

2. Draw the outline of the Venn diagram

Let X = the sample space; R = rugby; S = soccer and C = cricket. Put this information on a Venn diagram:



Calculate the unknown values:

Since 190 of the boys play at least one of the sports, using the values on our Venn diagram, we can set up the following equation to solve for x.

$$64+8+15+20+(x-15)+(70-x)+(55-x)=190$$

$$\therefore 217-x=190$$

$$\therefore x=27$$

a. Cricket and soccer but not rugby:

$$CS = x-15 = 27-15=12$$

$$CS = 12$$

b. We know the number of boys who play two or more of rugby, cricket or soccer and we know the total number of boys. Therefore, we can calculate the probability using the following equation:

$$P(\textit{At least Two Sports}) = \frac{n(RC) + n(RS) + n(CS) + n(RCS)}{n(X)}$$

$$= \frac{8 + 15 + 20 + 12}{200} = \frac{55}{200}$$

$$\frac{11}{40} = 0.275$$

2. In a class of 50 students 30 take philosophy 23 take Sociology and 12 take both. Find the probability that

- a. Takes Sociology**
- b. Both Sociology and Philosophy**
- c. Does not take Philosophy**
- d. Takes Sociology but not Philosophy**
- e. Takes Philosophy Given Sociology**
- f. Takes Philosophy or Sociology**
- g. Neither Philosophy Nor Sociology**
- h. Takes Sociology or Philosophy but not both**

Examples:

Q.1 The probability that the new rail budget will have an increase in the first class fare is 0.75. Increase in second class fare is 0.60 and increase in both the class fare is 0.45. Find the probability that the new rail budget will have an increase in

- i. Either first class or second class fare
- ii. Neither first class nor second class fare
- iii. Only first class fare

Q.2 In a group of 80 persons, 30 like Western Music, 40 like Indian music and 60 like at least one of the two types of music. Find the probability that person selected at random from this group

- i. Like both types of music
- ii. Likes only Indian Music
- iii. Does not like any of the type of music
- iv. Likes only one type of music

Tree Diagram

- It is useful for organising and visualising the different possible outcomes of a sequence of events.
- Each branch in the tree shows an outcome of an event, along with the probability of that outcome.
- For each possible outcome of the first event, we draw a line where we write down the probability of that outcome and the state of the world if that outcome happened.
- Then, for each possible outcome of the second event we do the same thing.
- The probability of a sequence of outcomes is calculated as the product of the probabilities along the branches of the sequence.

Examples:

1. A coin is flipped and a dice is rolled. What is the probability of getting a 'tail' and a '6'?
2. The probability of rainfall on Saturday is 0.46 . The probability of no rainfall on Sunday is 0.78. Draw a tree diagram and calculate the probability that it will rain on both days
3. The probability of it raining is 0.2. If it rains, the probability I am late is 0.3 and if it doesn't rain, the probability of me being late is 0.1.

Example

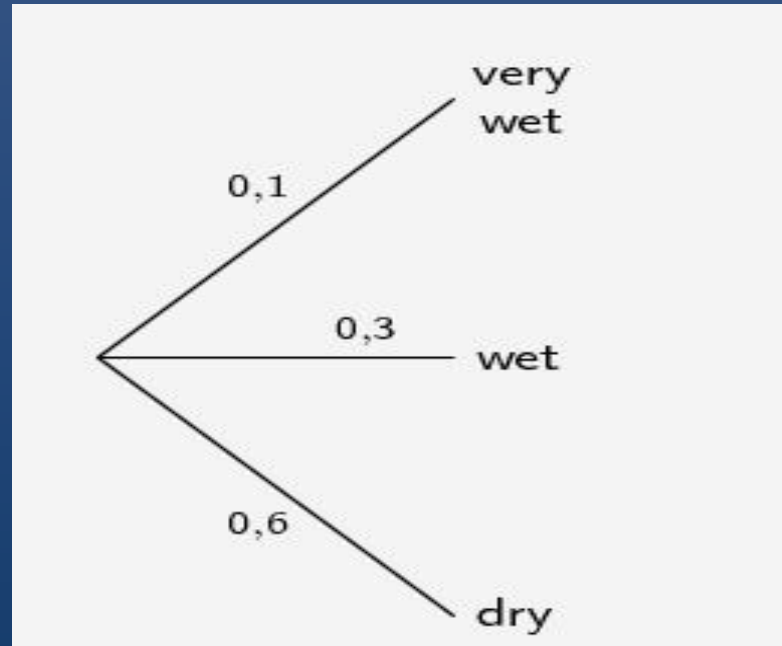
The probability that the floor of a supermarket will be wet when it opens in the morning is 30% and there is a 10% probability of the floor being very wet. The probability that a person will slip and fall if the floor is dry is 12% and a person is three times as likely to fall if the floor is wet. If the floor is very wet, the probability that a person will fall is 0.6. Draw a tree diagram to represent the given information, showing the probabilities of each outcome, and use it to answer the following questions:

- What is the probability that a person will fall on any given day?
- What is the probability that a person will not fall on any given day?
- Are the events of the floor being dry and a person falling independent? Justify your answer with a calculation.

Solution

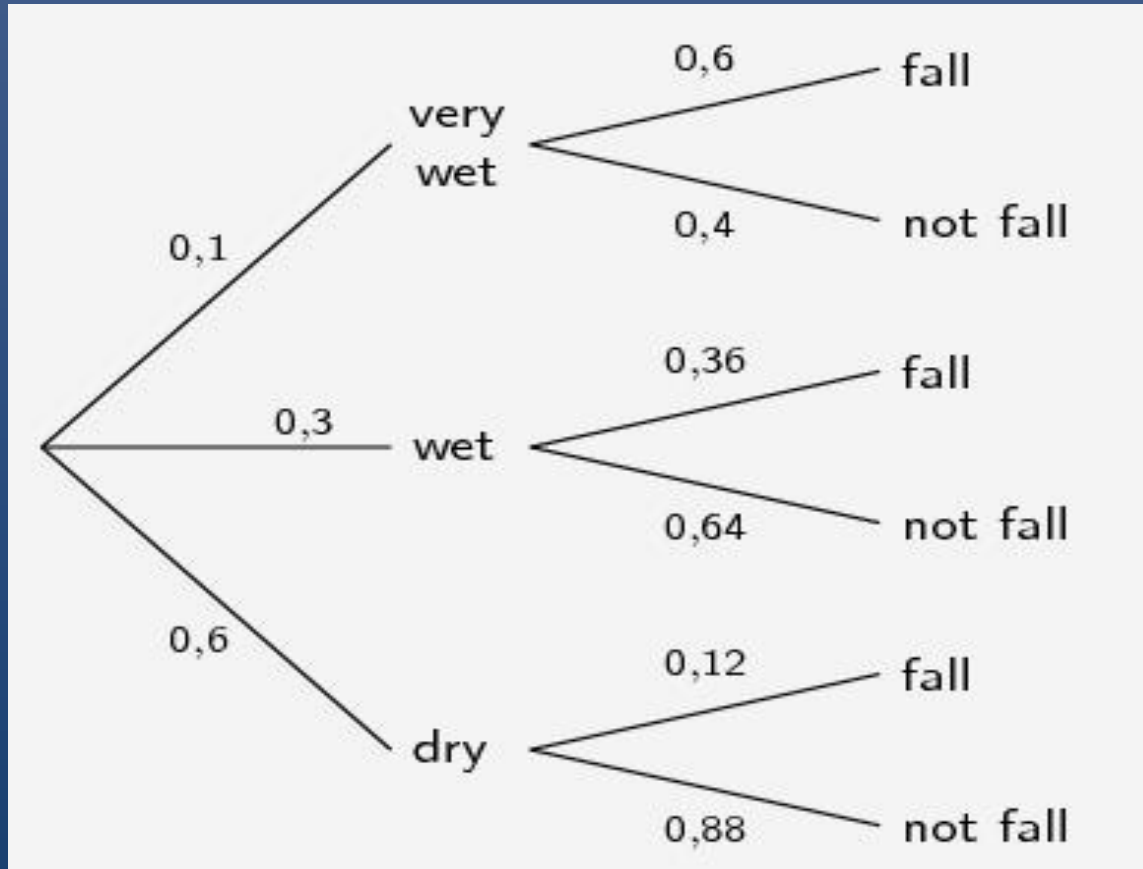
There are three outcomes for the floor, namely, dry, wet and very wet, and two outcomes for a person, namely fall or not fall.

Draw the first level of the tree diagram:



This tree diagram shows the possible outcomes and probabilities of the status of the floor.

Draw the second level of the tree diagram:



This tree diagram shows the possible outcomes and probabilities based on whether the floor is very wet, wet or dry. Remember that the sum of the probabilities for any set of branches is 1. Use this as a logical check whenever you are constructing a tree diagram.

a. We can calculate the probability of each outcome by multiplying the probabilities along the path from the start of the tree to the end of the branch containing the desired outcome.

$$P(\text{very wet and fall})=0.1 \times 0.6=0.06$$

$$P(\text{very wet and not fall})=0.1 \times 0.4=0.04$$

$$P(\text{wet and fall})=0.3 \times 0.36=0.108$$

$$P(\text{wet and not fall})=0.3 \times 0.64=0.192$$

$$P(\text{dry and fall})=0.6 \times 0.12=0.072$$

$$P(\text{dry and not fall})=0.6 \times 0.88=0.528$$

b. We can calculate the probability of falling or not falling by adding the probabilities of all the desired outcomes.

$$P(\text{fall})=0.06+0.108+0.072=0.24$$

$$P(\text{not fall})=0.04+0.192+0.528=0.76$$

Therefore the probability of falling on a given day is 24% and the probability of not falling is 76%.

c. Logically, it appears that these events are dependent but the question asked us to prove this using a calculation. We can do this using the rule for independent events:

$$P(A \text{ and } B) = P(A) \times P(B)$$

$$P(\text{dry and fall}) = 0.072$$

$$P(\text{dry}) \times P(\text{fall}) = 0.6 \times 0.24 = 0.144$$

$$\text{Therefore } P(\text{dry and fall}) \neq P(\text{dry}) \times P(\text{fall})$$

Therefore we can conclude that the floor being dry and a person falling are dependent events.

Two-way contingency tables

- These are a tool for keeping a record of the counts or percentages in a probability problem.
- Two-way contingency tables are especially helpful for figuring out whether events are dependent or independent.

Example:

The table below shows the results of testing two different treatments on 240 fruit trees which have a disease causing the trees to die. Treatment A involves the careful removal of infected branches and treatment B involves removing infected branches as well as spraying the tree with antibiotic.

	Tree dies within 4 years	Tree lives > 4 years	Total
Treatment A	70	50	
Treatment B			
Total	90	150	

Example Contd....

1. Fill in the missing values on the table.
2. What is the probability the a tree received treatment B?
3. What is the probability that a tree will live beyond 4 years?
4. What is the probability that a tree is given treatment B and lives beyond 4 years?
5. Of the trees who were given treatment B, what is the probability that a tree lives beyond 4 years?
6. Are a tree given treatment B and living beyond 4 years independent events? Justify your answer with a calculation.

Solution:

1. Complete the contingency table:

Since each column has to sum to its total, we can work out the number of trees which fall into each category for treatments A and B. Then, we can add each row to get the totals on the right hand side of the table.

	Tree dies within 4 years	Tree lives > 4 years	Total
Treatment A	70	50	120
Treatment B	20	100	120
Total	90	150	240

2. For the second question, we need to determine the probability that a tree receives treatment B. This means that we do not include treatment A in this calculation. So, the probability that treatment B is given to a tree is the ratio between the number of trees that received treatment B and the total number of trees.

$$P(\text{Treatment B}) = \frac{n(\text{treatment B})}{n(\text{total trees})} = \frac{120}{240} = \frac{1}{2}$$

3. The probability that a tree will live beyond 4 years:

$$\begin{aligned}P(\text{lives beyond 4 years}) &= \frac{n(\text{lives} > 4 \text{ years})}{n(\text{total trees})} \\&= \frac{150}{240} \\&= \frac{5}{8}\end{aligned}$$

4. The probability that a tree receives treatment B and lives beyond 4 years

$$\begin{aligned}P(\text{treatment B and lives} > 4 \text{ years}) &= \frac{n(\text{treatment B and lives} > 4 \text{ years})}{n(\text{total trees})} \\&= \frac{100}{240} \\&= \frac{5}{12}\end{aligned}$$

5. We need to determine the probability that of the trees which received treatment B, a tree lives beyond 4 years. This means we are only concerned with those trees which received treatment B. We no longer need to care about the trees given treatment A, so our denominator needs to be adjusted accordingly.

$$\begin{aligned} P(\text{lives} > 4 \text{ years having received treatment B}) &= \frac{n(\text{treatment B and lives} > 4 \text{ years})}{n(\text{total treatment B})} \\ &= \frac{100}{120} \\ &= \frac{5}{6} \end{aligned}$$

6. We need to determine whether a tree given treatment B and living beyond 4 years are dependent or independent events. According to the definition, two events are independent if and only if

$$P(A \text{ and } B) = P(A) \times P(B)$$

$$\begin{aligned} P(\text{treatment B}) \times P(\text{lives} > 4 \text{ years}) &= \frac{1}{2} \times \frac{5}{8} \\ &= \frac{5}{16} \end{aligned}$$

$$P(\text{treatment B and lives} > 4 \text{ years}) = \frac{5}{12}$$

From these probabilities we can see that;

$$P(\text{treatment B and lives} > 4 \text{ years}) \neq P(\text{treatment B}) \times P(\text{lives} > 4 \text{ years})$$

and therefore the treatment of a tree with treatment B and living beyond 44 years are dependent events.

Suppose a study of speeding violations and drivers who use cell phones produced the following fictional data:

	Speeding violation in the last year	No speeding violation in the last year	Total
Uses cell phone while driving	25	280	305
Does not use cell phone while driving	45	405	450
Total	70	685	755

- a. Find $P(\text{Driver is a cell phone user})$.
- b. Find $P(\text{Driver had no violation in the last year})$.
- c. Find $P(\text{Driver had no violation in the last year and was a cell phone user})$.
- d. Find $P(\text{Driver is a cell phone user or driver had no violation in the last year})$.
- e. Find $P(\text{Driver is a cell phone user given that driver had a violation in the last year})$.
- f. Find $P(\text{Driver had no violation last year given that driver was not a cell phone user})$.

Geometric Probability

- Geometric probability is the visual representation of probability.
- Many of the probabilities are continuous and for these probabilities, it is easy to represent as a geometric probability.
- The area of the geometric figure is used to compute the probabilities. The geometric probability is the probable area divided by the entire area.
- The probabilities which are non-discrete or are continuous are represented across a number line or across a two-dimensional plane.
- The experiment having numerous probability outcomes are often represented as a geometric probability.
- For calculation of geometric probability, we are only required to know the expected area where we want to hit with reference to the total area.
- Geometric probability is obtained by dividing the expected area by the total area.
- Geometric Probability = Probable Area/Total Area

Examples

- The probability of the arrival of a train across a time period is a continuous probability, which can be represented as a geometric probability.
- In a class, the probability of a number of students gaining 65% or more marks can be represented as a geometric probability. This can be represented as a normal distribution, and the probability region can be shaded.
- Among a group of people, 30% drink tea, 40% drink coffee, and the remaining 30% drink soft drinks. This can be represented as a shaded region in a pi diagram.
- The probability of incidence of covid-19 among the population can be graphically represented as a geometric probability.

Representation of Geometric Probability

Number Line:

- The continuous probabilities are marked as a point on the number lines, and the length of the line from the starting point divided by the entire length of the line gives the probability answer.

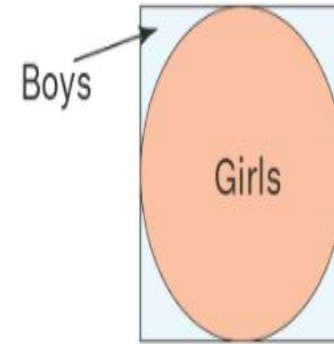
Two Dimensional Plane:

- The geometric probability can be represented in the two-dimensional graph as a normal curve, as a pi diagram, or any other geometric shape.
- For the representation of the probability of a single event we use the normal curve, and represent the probability as the part of the area of the normal curve.
- And for representing multiple probabilities we use a Pi diagram or any other diagram. Further, we can represent geometric probability in higher dimensions also.

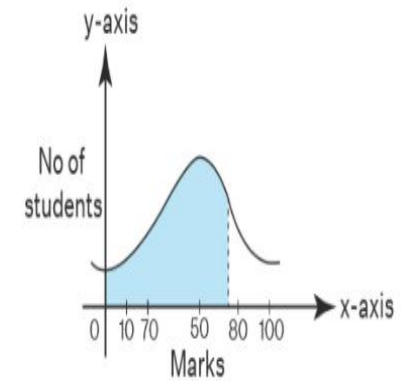
Percentage Format:

- Geometric probability is numerically represented as a percentage.
- The percentage value represents the happening of the event or the area of the geometric figure.
- 100% refers to the entire shaded image and it means the sure happening of the event.
- A 50% is represented as a half-shaded region and is equal to half of the chances of happening of an event.

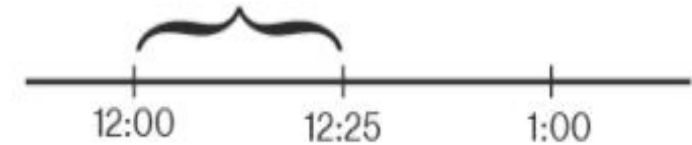
Representation of Geometric Probability



$$P(\text{Girls}) = \frac{\text{A of Circle}}{\text{A of Square}}$$



$$P(\text{Marks} < 70) = \frac{\text{Area of Shaded Region}}{\text{Total Area under the curve}}$$



$$P(\text{Bus arriving between 12:00-12:25}) = \frac{\text{Length 12:00 - 12:25}}{\text{Length 12:00 - 1:00}}$$

Example

Find the geometric probability of selecting a girl from a class, if the girls of the class are represented by a circle of radius 3.5 cms, and the entire class of students is represented by a larger square of side 8 cms.

Solution:

Given that the girls represent the circle of radius 3.5cms, and the entire class of students represent a square of side 8 cms.

$$\text{Area of the circle} = \pi r^2 = 22/7 \times 3.5 \times 3.5 = 38.5$$

$$\text{Area of the square} = s^2 = 8 \times 8 = 64$$

Probability of selecting the girl = Area of Circle/Area of Square

$$\text{Probability of selecting the girl} = 38.5/64$$

Therefore the probability of selecting the girl from the class of students is 38.5/64.

Take Home Task

Question :

Find the geometric probability of the number of students who can speak English, if the people are represented s sectors in a circle, and the English-speaking people represent a sector of 120° .

Solution:

The number of people who speak English is represented by a sector of angle 120° , and the total number of students of the class are represented by the entire circle, which is an angle of 360° .

Probability of people speaking English = Area of sector/Area of circle = Angle of the sector/Complete angle

Probability of people speaking English = $120^\circ/360^\circ = 1/3$

Therefore the probability of people speaking English language is $1/3$.